

This project aims to develop a smart pillbox using an Arduino and an MC-38 magnetic sensor to monitor whether the pillbox has been opened within a specified timeframe. If the pillbox remains unopened past the scheduled medication time, the system will trigger an alert (such as a buzzer, LED, or mobile notification) to remind the user to take their medication. By automating adherence tracking, this smart pillbox can help patients manage their medication schedules more effectively, improving health outcomes and reducing the risk of missed doses.

The science behind the effectiveness of taking your medication on time is undeniable. That is why we have a notification to remind the user to take their medication as the most important priority. It is shown that people nowadays are less inclined to write to-do lists, so providing them with a screen will acclimate to their daily life easier and remind them more effectively. Considering the mental state and ability of the user to remember to take their medication, we must assume that they are struggling with a chronic illness that affects their mental health and physical health immensely.

Our feedback from experts included not only having 7 spots for days of the week but also am and pm slots since some people take medications in the morning and at night, at different times of the day. The IT workers helped give us feedback on things we had not thought about previously. We struggled with connecting the wires to make it read and notify the phone. This is something we would need to address and fix if we had time to fix the lid of the pill box. So if we got back into fixing the issues to accommodate their requests, we would make it 7 across and each slot would have 2 rows for AM and PM. Then the alerts still occur if the lid is opened, but we would have to work out how the pill box knows what time they opened it to make sure it is the right time and the stakeholder doesn't miss their time to take their correct time.